

1. What is the main difference between a Python function and a SageMath function?
 - (a) For which problems is a Python function better than a SageMath function?
 - (b) For which problems is a SageMath function better than a Python function?

Illustrate your explanations with examples we have covered in the course.

2. Consider the plane curve defined by $x^2y - y^2 + 3x - 6 = 0$ and the point $(2, 0)$.

View y as a function of x and compute $\frac{dy}{dx}$. What is $y'(2) = \frac{dy}{dx}\bigg|_{x=2}$?

3. Let A be a symbolic n -by- n matrix, where the (i, j) -th entry $a_{i,j} = x_i^{j-1}$, for i and j ranging both from 1 to n .

Define a function F which takes on input a list X of n variables and which returns the determinant of A . Show that your function works for $n = 5$.

4. Denote by T the Taylor series $2 + t - 3t^2 + 2t^4 + O(t^5)$.

Compute the inverse T^{-1} and verify that T times T^{-1} equals $1 + O(t^5)$.

5. Write an efficient recursive function to compute

$$q_0(x) = 1, \quad q_1(x) = x + 1, \quad \text{for } n > 1 : q_n(x) = (2x - 1)q_{n-1}(x) + (x + 1)q_{n-2}(x).$$

Make sure your function can compute $q_{100}(x)$. Do **NOT** use `cached_function`.

6. Plot the curve defined by $x^4 + x^3y - y^2 = 0$, for a good range for x and y .

- (a) Convert the equation into polar coordinates and plot.
- (b) Describe the difference between the two plots.

7. Define $p = x(y^2 - xz)^2 - y^5$ and $q = y^4 + y^3z - x^2z^2$ as polynomials with rational coefficients, with a lexicographic monomial order $x > y > z$.

- (a) Use a resultant to eliminate x and to obtain a polynomial r in y and z .
Use a resultant to eliminate x and to obtain a polynomial s in x and z .
- (b) Factor r and use a polynomial factor to compute y as an expression of z .
Factor s and use a polynomial factor to compute x as an expression of z .
- (c) What can you state about the shape of the space curve defined by p and q ?

8. Consider $x(t)$ a function of t as the solution of the initial value problem

$$\frac{d^2}{dt^2} x(t) - \frac{d}{dt} x(t) - 2x(t) = 0, \quad x(0) = 1, \quad x'(0) = 0.$$

Compute $x(t)$ using the Laplace transform.

What is the algebraic equation that must be solved?

9. Let $P = [(5, 6), (9, 1), (4, 8), (2, 2), (2, 4), (3, 3), (1, 4), (4, 6)]$.

Compute the half planes that define the convex hull spanned by the points in P .